

Relativity from Relative Entropy

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Spurred by Einstein's appendix to *The Meaning of Relativity*, we embark on an algebraic voyage from von Neumann algebras to the emergence of special relativity from algebraic structure. We identify time with modular flow (in line with Connes and Rovelli's Thermal Time Hypothesis), space with a suitable commutative subalgebra, and observers themselves with cyclic, separating states. Comparisons between observers are governed by the Connes cocycle, with the real-time cocycle near $t = 0$ yielding a relative entropy S that we interpret as comparing the frequencies of their respective modular "clocks", and the Radon-Nikodym point $t = -i/2$ giving a fidelity $\mathcal{F}$ that compares the measure density of modular "rods". In a Type I Gibbs model, these quantities exactly obey $\mathcal{F}^2 = e^{-S}$ in the well-localized (single eigenvalue) limit. We then analyze the Rindler wedge, where the vacuum modular Hamiltonian generates Lorentz boosts, and discuss localized coherent excitations for which the relative entropy is computable in terms of boosts and the cocycle admits a Gaussian closed form at the Radon-Nikodym point. In this regime, special-relativistic kinematics emerges from modular data. We end with a survey of promising directions to continue voyaging.

1. Introduction

In the final appendix of the final edition of Einstein's *The Meaning of Relativity*, Einstein wrote:

One can give good reasons why reality cannot at all be represented by a continuous field. From the quantum phenomena it appears to follow with certainty that a finite system of finite energy can be completely described by a finite set of numbers (quantum numbers). This does not seem to be in accordance with a continuum theory and must lead to an attempt to find a purely algebraic theory for the representation of reality. But nobody knows how to find the basis for such a theory.

This may be the last thing the great scientist committed to print. It reflects a deep pessimism about the use of the continuum in modeling a quantum-mechanical reality, and more than a whiff of despair about the unified field theory he had been labouring over for decades. Written a few months before he died, it's tempting to frame this as a jeremiad of defeat. Maybe so. But it has a specificity suggesting Einstein knows more than he says; that perhaps he has been considering the very "algebraic theory" he decries as unknown. Not the mature fruit of scientific work, of course; that he would have published. But the man who spent so many hours wandering the byways of Princeton, lost in thought, might have devoted some of those hours to his final algebraic challenge. To borrow an image of Grothendieck,¹ the problem might remain unsolved, but Einstein's sea might have been risen to insensibly surround it.

Let us suppose, then, that Einstein spent the odd Princeton ramble pursuing his own hint, immersed in (or immersing) the problem of replacing the continuum with a theory in accord with the facts of quantum mechanics. What might it look like? The most obvious place to start would have been the work of Irving Segal, a former postdoc of Einstein's at the IAS, or his Princeton colleague

¹ "The unknown thing to be known appeared to me as some stretch of earth or hard marl, resisting penetration... The sea advances insensibly and in silence, nothing seems to happen, nothing moves, the water is so far off you hardly hear it... yet it finally surrounds the resistant substance." From *Récoltes et Semailles* (1986).

John von Neumann, both of whom approached algebra from the viewpoint of quantum-mechanical foundations.² Von Neumann (and his colleague Francis Murray) had defined objects called “rings of operators”, later to be known as *W*-algebras* or simply *von Neumann algebras*; Segal generalized these to a broader class of *C*-algebras*. The literature was (and is) highly technical, and one can easily imagine Einstein, fed up with mathematical jargon, concluding he was too old to learn new tricks and retreating to the *terra firma* of his unified field theory.

But it is possible things went differently. Von Neumann was still at Princeton and dwelling on old flames; his 1954 ICM address on unsolved problems³ exclusively concerned technical, physical, and logical/probabilistic aspects of von Neumann algebras. To the assembled mathematicians, it seemed like “warmed-up soup”, but it may have reflected genuine mathematical recidivism rather than a laziness uncharacteristic of von Neumann. So, it is not impossible that Einstein and von Neumann crossed paths in New Jersey in 1954; and that, amidst the Institute banter, polite avoidance of nuclear issues, and standard issue shop talk, they touched on a problem dear to both: the application of algebraic methods to a quantum-mechanical reality.

This essay is an attempt to honour this (perhaps) imaginary conversation between von Neumann and Einstein, and to trace out the few steps Einstein might have taken armed with his delicate sense of physical truth, von Neumann’s algebraic power, and (anachronistically) a slew of deep classification results that would arrive in the 60s and 70s. This anachronism is regrettably necessary because, without the physically suggestive character of these results, it is impossible to imagine Einstein taking interest in something as *recherché* as the taxonomy of infinite-dimensional algebras. Our proposal is speculative, informal, and (we hope) entertaining. Let us now embark on Einstein’s lost voyage.

2. Background

This section will outline standard algebraic background material, albeit in a slightly nontraditional way.

2.1. Von Neumann algebras

A *von Neumann algebra* is a collection $\mathfrak{M}$ of operators which, roughly speaking, correspond to measurements an observer can make, actions they can perform, or equivalently, transitions between states of the observer. This means that:

- if $a \in \mathfrak{M}$ and $\lambda \in \mathbb{C}$, then $\lambda a \in \mathfrak{M}$ since we can scale measurements;
- if $a, b \in \mathfrak{M}$, then $a + b \in \mathfrak{M}$ since we can add measurements;
- if $a, b \in \mathfrak{M}$, then $a \cdot b \in \mathfrak{M}$ since we can compose transformations;
- there is an antilinear operator $*$: $\mathfrak{M} \rightarrow \mathfrak{M}$ satisfying $(ab)^* = b^*a^*$, with a^* meaning we apply to the final state; and
- there is a norm $\| \cdot \| : \mathfrak{M} \rightarrow \mathbb{R}_{\geq 0}$ satisfying $\|a\|^2 = |\lambda_{\max}| \in \mathbb{R}_{\geq 0}$ is a norm, where $\lambda_{\max}$ is the largest eigenvalue of a^*a , since measurements are finite.

² “Irreducible representations of operator algebras” and “Postulates for General Quantum Mechanics” (1947), Irving Segal; “Rings of Operators I” (1936) and “Rings of Operators II” (1937), Murray and von Neumann.

³ “Unsolved Problems in Mathematics” (1954), International Congress of Mathematics. The “soup” comment comes to us from Freeman Dyson via George Dyson, *Turing’s Cathedral* (2012).

So far, this defines a C^* -algebra (once we supplement with standard axioms, e.g. $\mathfrak{M}$ is a vector space, the product is associative, etc, which we omit). The notion of a transition between states requires elaboration. Usually, a state $\phi \in \mathfrak{M}_+^*$ is viewed as a linear functional $\phi : \mathfrak{M} \rightarrow \mathbb{C}$ (written $\phi \in \mathfrak{M}^*$) which is positive, in that $\phi(a^*a) \geq 0$ (written $\phi \in \mathfrak{M}_+^*$) and normalized so that $\phi(e) = 1$, where $e \in \mathfrak{M}$ is the identity. We denote Napier's constant by e .

Given a state, we can build a Hilbert space as follows: define the “sesquilinear” form $\langle a, b \rangle = \phi(a^*b)$, quotient out the null space $\mathcal{N}_\phi$, and define $\mathcal{H}_\phi$ as the vector space formed by equivalence classes $\mathfrak{M}/\mathcal{N}_\phi$, completed with respect to the inner product

$$\langle \psi | \omega \rangle = \langle [a]_\phi, [b]_\phi \rangle = \phi(a^*b) \quad (1)$$

where the elements $\psi = [a]_\phi, \omega = [b]_\phi \in \mathcal{H}_\phi$. If the state ϕ is *faithful*, meaning that $\phi(a) = 0$ implies $a = 0$ and hence the null space is trivial $\mathcal{N}_\phi = \{0\}$, the induced Hilbert space $\mathcal{H}_\phi$ faithfully embeds $\mathfrak{M} \subseteq \mathfrak{B}(\mathcal{H})_\phi$. We always think of a von Neumann algebra “concretely”, that is, as a subalgebra of bounded operators acting on some Hilbert space, $\mathfrak{M} \subseteq \mathfrak{B}(\mathcal{H})$.

We can view the inner product $\langle \psi | \omega \rangle$ as computing a *transition amplitude* from $|\omega\rangle$ to $|\psi\rangle$. This gives the concrete interpretation of a^* as applying a to the final state, since

$$\langle \psi | a | \omega \rangle = \langle \psi | a \omega \rangle, \quad \langle \psi | a^* | \omega \rangle = \langle a \psi | \omega \rangle. \quad (2)$$

Now we come to the crux: in addition to the axioms above, a von Neumann is closed in the *weak operator topology*,⁴ meaning that for any convergent sequence of operators $a_n \rightarrow a$, the limit

$$\langle \psi | a_n | \omega \rangle \rightarrow \langle \psi | a | \omega \rangle \quad (3)$$

for any pair of vectors $|\psi\rangle$ and $|\omega\rangle$. More plainly, transition amplitudes are a continuous function of the transition operator a . Physically, this corresponds to the ability to *approximate transition amplitudes*, a reasonable constraint for observers limited to finite-precision measurements.⁵

2.2. Space

As a starting point for a model of reality, an algebra is strikingly minimal. We have no space, no time, nothing to pin the theory to the suspect continuum; just a few axioms about observation thrown together. Whether these are the right axioms, or some other flavour of algebraic object, is a matter of debate. But we will see that the structure of von Neumann algebras allows for something remarkable: the *emergence* of structure resembling space and time.

Let's start with space. Suppose for a moment that observables in $\mathfrak{M}$ are labelled by location in time and space, $a(\mathbf{x}, t)$. Two observables $a(t, \mathbf{x}), b(t, \mathbf{x}')$ sharing the same time, but differing in place, should not be able to influence each other, either as measurements or actions; this would correspond to instantaneous transmission of information. A robust feature of nature is that it is *local*, i.e. no instantaneous transmission allowed. In terms of transition amplitudes, this means that the order we apply them in should be immaterial,

$$\langle \psi | a(t, \mathbf{x}) b(t, \mathbf{x}') | \omega \rangle = \langle \psi | b(t, \mathbf{x}') a(t, \mathbf{x}) | \omega \rangle \quad (4)$$

⁴ Called “weak” since it is smaller than another standard operator topology, known in the trade as the “strong operator topology”.

⁵ For a thoroughgoing introduction to von Neumann algebras, see *Fundamentals of the Theory of Operator Algebras: Advanced Theory* (1997), Kadison and Ringrose.

for any initial and final states, or equivalently, $ab - ba = [a, b] = 0$, so a and b commute. On the other hand, if it is possible to send a message between two spacetime points $(t, \mathbf{x})$ and $(t', \mathbf{x}')$, and it is possible for the observer to distinguish the excitation, we can construct initial and final states such that

$$\langle \psi | a(t, \mathbf{x}) b(t', \mathbf{x}') | \omega \rangle \neq \langle \psi | b(t', \mathbf{x}') a(t, \mathbf{x}) | \omega \rangle, \quad (6)$$

and hence $[a, b] \neq 0$. The final state simply records which operator was observed first! We say that points which can message each other are *timelike-separated* and points that cannot are *spacelike-separated*. Operators at spacelike separated points commute. This provides a simple algebraic recipe for space: we consider a subalgebra $\Sigma \subseteq \mathfrak{M}$ of commuting operators that cannot be enlarged, also called a *maximal abelian subalgebra (masa)*, since this roughly corresponds to a Cauchy slice. Since observers can be localized to regions of spacetime with associated *accessible* von Neumann subalgebra $\mathfrak{M}_{\text{obs}} \subseteq \mathfrak{M}$, we will weaken this to possibly observer-dependent masas $\Sigma_{\text{obs}} \subseteq \mathfrak{M}_{\text{obs}}$ (Fig. 1).

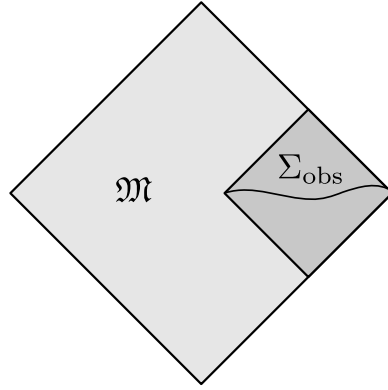


Figure 1: A global algebra $\mathfrak{M}$ containing a subalgebra $\mathfrak{M}_{\text{obs}}$ accessible to an observer, which in turn contains a masa Σ_{obs} .

This does not immediately give space as we usually understand it, that is, via coordinates. To induce coordinates, we use a beautiful result called the *Gelfand isomorphism*. This associates a commutative C^* -algebra Σ to its set of states, called in this context the *spectrum* $X_{\text{obs}} = \text{spec}(\Sigma_{\text{obs}}) \subseteq \Sigma_{\text{obs}}^*$. These give rise to coordinates in the sense that Σ can simply be viewed as continuous functions on X_{obs} , with

$$\Sigma_{\text{obs}} \cong C^0(X_{\text{obs}}) \quad (7)$$

as a norm-preserving isomorphism.⁶ Concretely, the spaces are identified by the *Gelfand map* $\Phi : \Sigma_{\text{obs}} \rightarrow X_{\text{obs}}$, whose seemingly innocuous definition

$$\Phi(a)(\phi) = \phi(a) \quad (8)$$

underlies Fourier series ($\Sigma = S^1$), the Fourier transform ($\Sigma = \mathbb{C}$), and the Mellin transform ($\Sigma = \mathbb{R}_{>0}^\times$) among others! For instance, on the circle $\Sigma = S^1 = [0, 2\pi)$ (with endpoints identified), the characters are the exponential maps

$$\phi_k(z) = e^{2\pi i k z} \quad (9)$$

for $k \in \mathbb{Z}$. Thus, we can view $\text{spec}(S^1) = \mathbb{Z}$ as the integers labelling the familiar Fourier modes on a circle.

⁶ With respect to the uniform norm on functions. For details, see *Introduction to Topology and Modern Analysis* (1964), George Simmons.

Hence, to describe space algebraically, we therefore first identify an accessible masa Σ_{obs} , and arrive at coordinates $X_{\text{obs}} = \text{spec}(\Sigma_{\text{obs}})$ given by the states on the algebra. The algebra will have *projections*

$$\Pi^2 = \Pi^* \Pi = \Pi \quad (10)$$

which we can view as indicator functions on their *support*:

$$\text{supp}(\Pi) = \{x \in X_{\text{obs}} : \Pi(x) \neq 0\}. \quad (11)$$

This allows us to talk of regions of space $R = \text{supp}(\Pi) \subset X_{\text{obs}}$; there is no guarantee we can talk about points (it depends on the projection structure of the masa) so this notion of space is already *non-local*. Given the difficulty of covariantly specifying a point in quantum gravity, i.e. without coordinates, this is a feature rather than a bug! The structure of a masa in a von Neumann algebra is highly constrained, with three basic flavours:⁷

- X finite, with model space $X \cong \{1, 2, \dots, n\}$;
- X infinite and discrete, with model space $X \cong \mathbb{N}$; and
- X continuous, with $X \cong [0, 1]$.

⁷ "Extensions of pure states" (1959), Kadison and Singer.

We can also take direct sums of these basic types. Only the first two allow us to project onto a point; thus, the price of having a continuum is non-locality. Einstein was right to be suspicious!

2.3. Time

Space is built out of the sharp notion of non-causation between oints. In contrast, there is no natural way to define a maximal set of non-commuting operators, since influence is a much richer notion than non-influence. To make matters worse, time has the puzzling phenomenological feature that (like the Marcusian man) it is *one-dimensional*. Finally, there should be a connection between thermodynamics and temporal flow, if we are to believe Eddington's thesis on the Arrow of Time.⁸ How can we reconcile these requirements?

Recall the interpretation of a^* : it is a applied to a final state rather than an initial state. Thus, given a unit vector $|\psi\rangle$ in the (faithful) Hilbert space $\mathcal{H}$, we already have a notion of earlier and later in time, and can ferret out asymmetries by considering how a^* affects the state in contrast to a . To probe this, it is not enough to consider

$$\langle \psi | a | \psi \rangle \quad \text{vs} \quad \langle \psi | a^* | \psi \rangle; \quad (12)$$

these are complex conjugates and classical observers only have access to probabilities rather than amplitudes, so they will see the same thing. The next simplest objects to consider are

$$\langle \psi | a^* a | \psi \rangle \quad \text{vs} \quad \langle \psi | a a^* | \psi \rangle. \quad (13)$$

When a is self-adjoint ($a^* = a$) or normal ($a^* a = a a^*$) these are trivially related, but for a general operator there is no immediate connection. To see why this

⁸ *The Nature of the Physical World* (1928), Arthur Eddington.

might be useful, consider the case of a harmonic oscillator with creation operator a^* and annihilation operator a .⁹ We note first that $N = a^*a$ is the number operator counting excitations, and second, a and a^* anticommute, so the difference obeys

$$\langle \psi | a^* a | \psi \rangle - \langle \psi | a a^* | \psi \rangle = 2\langle N \rangle_\psi, \quad (14)$$

i.e. twice the average number of particles in state $|\psi\rangle$. This example pleasingly illustrates that, in contrast to the commuting spatial case, time does appear to be tied to anticommutation, albeit between operators and their adjoints.

Let us try to capture the asymmetry more precisely. Consider the *modular operator* Δ_ψ , which swaps the adjoint operation under expectations for $|\psi\rangle$:

$$\langle \psi | a^* \Delta_\psi b | \psi \rangle = \langle \psi | a b^* | \psi \rangle \quad (15)$$

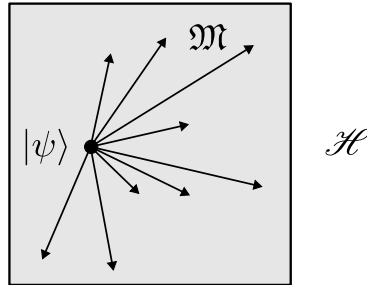
for all $a, b \in \mathfrak{M}$. Let us specialize to the case where the unit vector $|\psi\rangle$ is *cyclic*, meaning that $\mathfrak{M}|\psi\rangle$ is dense in the Hilbert space $\mathcal{H}$ that $\mathfrak{M}$ acts on (Fig. 2):

$$\overline{\mathfrak{M}|\psi\rangle} = \mathcal{H}. \quad (16)$$

Since $|\psi\rangle$ is dense in the full Hilbert space $\mathcal{H}$, and

$$\langle a\psi | \Delta_\psi | a\psi \rangle = \langle \psi | a a^* | \psi \rangle \geq 0, \quad (17)$$

we deduce that the modular operator is positive. Here, we are making tacit use of the weak closure property to densely approximate the full Hilbert space. Since Δ_ψ is positive, it is diagonalizable¹⁰ with positive eigenvalues $e^{-\lambda}$; these are precisely the operator asymmetries for the corresponding eigenvectors $a_\lambda |\psi\rangle$. Thus, if the map Δ_ψ exists, it tells us how “wonky” each operator is with respect to swapping initial and final states. Equivalently, it is the ability of a state to tell the time from comparing observables to their adjoints.



⁹ Technically, this isn't quite a von Neumann algebra, but we shouldn't let technical niceties get in the way of a good story. The correct treatment uses the CCR algebra, taming unbounded ladder operators by exponentiation.

¹⁰ We choose the form of eigenvalues so that $\lambda \in \mathbb{R}$, and bounded below just in case the modular operator is bounded.

Figure 2: The image of a cyclic vector $|\psi\rangle$ under algebra $\mathfrak{M}$ is dense in the full Hilbert space.

So far, we have only the discrete choice of initial or final state. From this binary, we would like to devise a continuous, one-dimensional flow of time. This seems quixotic, but if Δ_ψ is bounded, then the positive eigenvalues $e^{-\lambda}$ are bounded below, and hence the values λ are bounded below. This suggests we can view them as energy levels of what is called the *modular Hamiltonian* K_ψ , defined by

$$\Delta_\psi = e^{-K_\psi}. \quad (18)$$

If $|\psi\rangle$ is also *separating*, meaning $a|\psi\rangle = 0$ implies $a = 0$, then $\lambda = \infty$ is not an eigenvalue and K_ψ is itself bounded; this will prove to be a crucial technical

assumption. Time evolution with respect to the modular Hamiltonian is given by a propagator

$$e^{-itK_\psi} = \Delta_\psi^{it}. \quad (19)$$

Since the state is naturally fixed, we instead consider the Heisenberg picture evolution of an observable:

$$a \mapsto \Delta_\psi^{-it} a \Delta_\psi^{it} = \sigma_\psi^t(a). \quad (20)$$

Note that Δ_ψ is generically *not* contained in $\mathfrak{M}$, but operates on the larger Hilbert space $\mathcal{H}$. But a series of deep theorems proved by Tomita and championed by Takesaki¹¹ imply that, not only does this Heisenberg evolution stay inside the algebra, $\sigma_\psi^t(\mathfrak{M}) \subseteq \mathfrak{M}$, but represents a 1-parameter family of automorphisms of $\mathfrak{M}$, i.e. $\sigma_\psi^t : \mathfrak{M} \rightarrow \mathfrak{M}$ is an automorphism for any $t \in \mathbb{R}$. We will assume that the accessible subalgebra associated to an observer $|\psi\rangle$, call it $\mathfrak{M}_\psi$, is closed under the associated modular flow (Fig. 3).

¹¹ “On canonical forms of von Neumann algebras” (1967), Minoru Tomita; *Tomita’s theory of modular Hilbert algebras and its applications* (1970), Masamichi Takesaki.

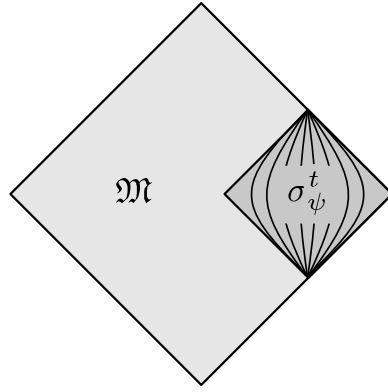


Figure 3: Modular flow σ_ψ^t within the accessible algebra.

Under modular flow, observables do not “leak out” of $\mathfrak{M}$ and into the ambient space of bounded operators $\mathfrak{B}(\mathcal{H})$; instead, they are quietly rearranged, akin to unitary time evolution in ordinary quantum mechanics. This suggests that modular flow might be identified with physical time, a proposal called the *Thermal Time Hypothesis* made by Connes and Rovelli.¹² Although this proposal has many merits, it is poor in concrete predictions. To learn what it means, we need to move into a bigger universe.

¹² “Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories” (1994), Alain Connes and Carlo Rovelli.

2.4. Observers

A lone observer makes for a dull cosmology. Introducing a second observer corresponds to introducing a second (cyclic, separating) state $|\omega\rangle$. The *relative modular operator* $\Delta_{\omega|\psi}$ is defined by

$$\langle \omega | a^* \Delta_{\omega|\psi} b | \omega \rangle = \langle \psi | a b^* | \psi \rangle, \quad (21)$$

so it not only swaps adjoints, but changes the reference state. As a simple example, $\Delta_{\omega|\psi}$ converts the expectation of a self-adjoint observable $b^* = b$ in state $|\omega\rangle$ to an expectation in state $|\psi\rangle$:

$$\langle \omega | \Delta_{\omega|\psi} b | \omega \rangle = \langle \psi | b^* | \psi \rangle = \langle \psi | b | \psi \rangle. \quad (22)$$

Again, $\Delta_{\omega|\psi}$ is positive with eigenvalues $e^{-\lambda}$ for bounded λ , giving a bounded *relative modular Hamiltonian* $K_{\omega|\psi} = -\log \Delta_{\omega|\psi}$. Morally, K_{ψ} captures the speed at which operators are moving when flipped relative to ψ , while $K_{\omega|\psi}$ captures how fast they are moving when flipped between states.¹³ It can be shown that the *relative modular flow* is identical to the modular flow induced by ω :

$$\sigma_{\omega|\psi}^t(a) = \Delta_{\omega|\psi}^{-it} a \Delta_{\omega|\psi}^{it} = \Delta_{\omega}^{-it} a \Delta_{\omega}^{it} = \sigma_{\omega}^t(a). \quad (23)$$

Formally, we can transition from the flow $\sigma_{\omega|\psi}^t = \sigma_{\omega}^t$ to the flow σ_{ψ}^t by conjugating with the *Connes cocycle*, defined by

$$[D\omega : D\psi]_t := \Delta_{\omega|\psi}^{-it} \Delta_{\psi}^{it} = \Delta_{\omega}^{-it} \Delta_{\psi}^{it}, \quad \sigma_{\psi}^t = \text{Ad}_{[D\omega : D\psi]_t} \circ \sigma_{\omega}^t. \quad (24)$$

Both $\Delta_{\omega|\psi}$ and Δ_{ψ} live in the larger algebra $\mathfrak{B}(\mathcal{H})$ rather than $\mathfrak{M}$ itself, but Connes showed that the parts outside $\mathfrak{M}$ always “cancel”, so (24) is in $\mathfrak{M}$. This result, called *Connes’ Cocycle Theorem*,¹⁴ is physically deep; it tells us that, for any two (suitably well-behaved) states, the modular flows they induce are equivalent up to an operator in $\mathfrak{M}$. This is analogous to timelike trajectories, which may differ in their precise bearing, but always point into the forward lightcone.

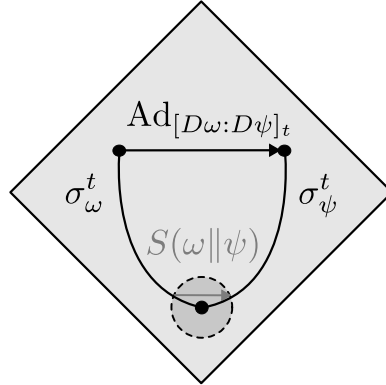
This raises a question: if space is defined by a commuting algebra Σ , are modular flows σ_{ψ}^t “timelike”, in the sense they point away from Σ ? More precisely, do operators generically fail to commute with their modular evolutions?

$$[a, \sigma_{\psi}^t(a)] \stackrel{?}{\neq} 0. \quad (25)$$

A short calculation shows that

$$\left. \frac{d}{dt} [a, \sigma_{\psi}^t(a)] \right|_{t=0} = [a, [iK_{\psi}, a]], \quad (26)$$

so commuting with modular flow means commuting with the modular Hamiltonian. In the finite-dimensional case, K_{ψ} is a matrix, and we can easily build non-commuting operators a which are not diagonal in the modular energy basis. This construction remains valid (with some bells and whistles) into infinite dimensions, so provided the modular Hamiltonian is not proportional to the identity $K_{\psi} \propto e$, modular flows are timelike in the sense described.



¹³ If this sounds like nonsense, try substituting a large eigenvalue into the defining equation.

¹⁴ *Noncommutative Geometry* (1990), Alain Connes.

Figure 4: The macroscopic and microscopic behaviour of the Connes cocycle: at arbitrary t , it changes σ_{ω}^t to σ_{ψ}^t ; at small t , it gives relative entropy.

To determine if the modular trajectory is timelike, we zoomed in on its behaviour at $t = 0$. We can perform a similar trick with two distinct observers (Fig.

4), $|\psi\rangle$ and $|\omega\rangle$, albeit with respect to the cocycle. If $|\psi\rangle = |\omega\rangle$, then

$$[D\omega : D\psi]_t = \Delta_\omega^{-it} \Delta_\psi^{it} = \Delta_\psi^{-it} \Delta_\psi^{it} = e \quad (27)$$

for all $t \in \mathbb{R}$. We might hope that the converse holds, and departures from identity are encoded at small t . To convert an operator into a scalar, we can just plug it into one of the states, say $|\psi\rangle$. Differentiating and taking $t = 0$ gives what is called the *relative entropy*

$$S(\omega||\psi) = i \frac{d}{dt} \langle \psi | [D\omega : D\psi]_t | \psi \rangle \Big|_{t=0}, \quad (28)$$

where the constant is chosen so that¹⁵

$$S(\omega||\psi) = \langle \psi | \left(\log \Delta_\psi - \log \Delta_{\omega|\psi} \right) | \psi \rangle. \quad (29)$$

¹⁵ There are higher-order terms in t by the Baker-Campbell-Hausdorff formula, but these disappear at $t = 0$.

Assuming that $|\psi\rangle$ is normalized, the term

$$\langle \psi | \log \Delta_\psi | \psi \rangle = \log(\langle \psi | \Delta_\psi | \psi \rangle) = \log(\langle \psi | \psi \rangle) = 0 \quad (30)$$

This leaves the Araki form of relative entropy¹⁶ $S(\omega||\psi) = -\langle \psi | \log \Delta_{\omega|\psi} | \psi \rangle$. Substituting the real inequality $\log x \leq x - 1$, we get an operator inequality with lower bound

$$S(\omega||\psi) \geq 1 - \langle \psi | \Delta_{\omega|\psi} | \psi \rangle = 1 - \langle \omega | \omega \rangle = 0. \quad (31)$$

¹⁶ “Relative entropy of states of von Neumann algebras” (1976), Huzihiro Araki.

It follows that relative entropy is always non-negative. With a little more work, one can show that $S(\omega||\psi) = 0$ just in case $\omega = \psi$, so cocycle behaviour in the vicinity of $t = 0$ precisely encodes when states are the same or different.

3. Modular relativity

Now that we have the ingredients, we can ask how our definitions of space, time, and observers fit together. We will see that they can be coherently assembled into a theory of *modular relativity*, extending the Thermal Time Hypothesis from a single global modular flow, to states-as-observers with compatible flows and nontrivial interactions.

3.1. Clocks

We start by recasting the relative entropy in a suggestive form. We have been associating modular flow to Hamiltonians of various sorts, which posit eigenvalues as scaling factors (for real time) or frequencies (for imaginary time). We can perform the same trick with the cocycle itself, although it has a slightly different structure. Define the *cocycle Hamiltonian* by $[D\omega : D\psi]_t = e^{-itH}$, so

$$\varphi(t) = \langle \psi | [D\omega : D\psi]_t | \psi \rangle = \langle e^{-itH} \rangle_\psi \quad (32)$$

is the *characteristic function* of $-H$ in state ω . Expanding in eigenfrequencies of H , we obtain the expression

$$\varphi(t) = \int e^{-i\lambda t} d\mu(\lambda) = 1 - it \int \lambda d\mu(\lambda) + \mathcal{O}(t^2). \quad (33)$$

By definition, $\varphi'(0) = iS(\omega\|\psi)$, and we learn that relative entropy equals average frequency of the cocycle:

$$S(\omega\|\psi) = \int \lambda \, d\mu(\lambda). \quad (1)$$

Positivity of relative entropy¹⁷ then implies an interesting fact: the mean cocycle frequency is always positive, and vanishes just in case the states coincide.

¹⁷ Here, positivity is a euphemism for non-negativity.

$$\mathcal{L}_{H_0, H_1}^\phi(t) = \langle \phi | \begin{array}{c} \xrightarrow{e^{-iH_1 t}} \\ \xleftarrow{e^{iH_0 t}} \end{array} | \phi \rangle$$

Figure 5: The Loschmidt echo: evolve $|\phi\rangle$ by a Hamiltonian H_1 forwards for time t , then backwards with H_0 for time t , and overlap with $|\phi\rangle$.

We can make this picture more physical using the notion of the *Loschmidt echo*.¹⁸ In general, we can compare two Hamiltonians H_0 and H_1 on a reference state $|\phi\rangle$ by evolving forward a time t with H_1 , then backward with H_0 , and taking the overlap with $|\phi\rangle$:

$$\mathcal{L}_{H_0, H_1}^\phi(t) = \langle \phi | e^{itH_0} e^{-itH_1} | \phi \rangle. \quad (34)$$

The difference between Hamiltonian spectra will manifest as *dephasing* and the loss of overlap. Typically, the focus is on the Loschmidt probability, i.e. the *squared amplitude* $|\mathcal{L}_{H_0, H_1}^\psi(t)|^2$. This automatically cancels the first order term and is guaranteed to be non-increasing at second order because $t = 0$ is a maximum:

$$\frac{1}{t^2} (1 - |\mathcal{L}_{H_0, H_1}^\psi(t)|^2) + \mathcal{O}(t^2) \geq 0. \quad (35)$$

We are more interested in the linear term, whose sign is not constrained in general. To understand what it signifies, consider a simple situation where $|\psi\rangle$ is an eigenvector of the two Hamiltonians, with eigenvalue λ_0, λ_1 respectively. Then

$$\mathcal{L}_{H_0, H_1}^\psi(t) = e^{-it(\lambda_1 - \lambda_0)} = 1 - it(\lambda_1 - \lambda_0) + \mathcal{O}(t^2), \quad (36)$$

so the Loschmidt echo records the frequency difference at first order. In plain language, it tells us which Hamiltonian *runs faster*. In general, we can use Baker-Campbell-Hausdorff to write

$$e^{itH_0} e^{-itH_1} = \left(1 + \mathcal{O}(t^2)\right) e^{-it(H_1 - H_0)} = \left(1 + \mathcal{O}(t^2)\right) \int e^{-it\lambda} \, d\lambda, \quad (37)$$

where we integrate over the spectrum of $H_1 - H_0$. Expanding $|\phi\rangle$ in eigenstates $a_\lambda|\phi\rangle$ (taking $|\phi\rangle$ cyclic), with coefficients c_λ , gives

$$\mathcal{L}_{H_0, H_1}^\psi(t) = \left(1 + \mathcal{O}(t^2)\right) \int e^{-it\lambda} c_\lambda \langle \phi | a_\lambda | \phi \rangle \, d\lambda. \quad (38)$$

This tells us which Hamiltonian runs faster *on average*, as measured by $|\phi\rangle$ in the sense that it is weighted by $d\mu(\lambda) = c_\lambda \langle \psi | a_\lambda | \psi \rangle \, d\lambda$, i.e. how much each eigenvector contributes to $|\phi\rangle$. For a first order term $-it\Delta$, H_1 runs faster for $\Delta > 0$ and H_0 runs faster for $\Delta < 0$.

¹⁸ Named after (but not by) Johann Josef Loschmidt, who took Ludwig Boltzmann to task over issues of reversibility in *classical* physics. See “Dynamics of Loschmidt echoes and fidelity decay” (2006), Thomas Gorin, Tomaz Prosen, Thomas H. Seligman, and Marko Znidaric for a thorough review.

This leads us to a second interpretation of the characteristic function $\varphi(t)$ as a Loschmidt echo between modular flows for $|\omega\rangle$ and $|\psi\rangle$ respectively. In this case, we are evolving *backward* in time t first (due to our “physics-style” convention for conjugating operators) with reference state $|\psi\rangle$, $H_0 = K_\psi$ and $H_1 = K_{\omega|\psi}$. Since $\sigma_{\omega|\psi}^t = \sigma_\omega^t$, we can simplify by replacing $K_{\omega|\psi}$ with K_ω in the cocycle structure, which yields

$$\varphi(t) = \mathcal{L}_{K_\psi, K_\omega}^\omega(-t). \quad (39)$$

The fact we first evolve backwards implies that, for first order term $-it\Delta$, K_ψ runs faster for $\Delta < 0$ and K_ω runs faster for $\Delta > 0$.

Positivity of relative entropy then implies a remarkable conclusion: a state $|\psi\rangle$ measures its own modular evolution to run faster, on average, than any other state. This is what it means for the mean cocycle frequency to be positive. This should of course remind of something else: a clock always measures its own time to run faster than any clock moving relative to it, an effect known as *time dilation*. As with regular time dilation, the puzzling symmetry of modular time dilation is resolved by observer-dependence: observer $|\omega\rangle$ measures $|\psi\rangle$ to run slower, and vice versa, because measurement itself breaks the symmetry.

Time dilation is a cornerstone of Einstein’s theory of special relativity. The existence of a modular analogue raises three natural questions:

- Is there an invariance principle (like constancy of c) grounding this effect?
- Is there a modular length contraction principle?
- Finally, how does modular time dilation relate to normal time dilation?

We will answer these questions in turn. The first is the simplest. The speed of light can be measured by, say, bouncing a laser off a mirror. The invariance of c means that the result of this experiment is the same whether the mirror is stationary or moving, or equivalently, how we boost relative to it. These boosts are *symmetries of the vacuum*, so in effect, this says that if we asymmetrically apply a symmetry of the vacuum, experiments determining c are invariant.

In the modular case, we treat the state $|\psi\rangle$ as the vacuum and $|\omega\rangle$ as the experimenter. Instead a change of frame, we will have a *-automorphism $\alpha : \mathfrak{M} \rightarrow \mathfrak{M}$ which relabels elements. This is a symmetry of $|\psi\rangle$ just in case

$$\psi \circ \alpha(a) = \langle \psi | \alpha(a) | \psi \rangle = \langle \psi | a | \psi \rangle = \psi(a). \quad (40)$$

Instead of computing the speed of a laser, the simplest thing we can compute is the relative entropy itself, $S(\omega || \psi)$. Asymmetrically “boosting” the experimenter means applying the automorphism α to their frame. This gives

$$S(\omega \circ \alpha || \psi) = S(\omega || \psi \circ \alpha), \quad (41)$$

using the symmetry (40). But applying an automorphism to *both* reference states cannot change the relative entropy; it relabels the eigenvectors of the cocycle Hamiltonian but leaves the spectrum invariant. Thus, we learn that relative entropy is the invariant grounding modular time dilation.

3.2. Rods

Next, we turn to length contraction. This is an intrinsically trickier problem; clocks are localized so a comparison is direct, whereas volumes are spread out. The trick is to look at the volume *density*, and happily there is an analogue of the relative entropy for comparing these. Instead of evaluating the cocycle at real time $t \in \mathbb{R}$, we look at a specific imaginary time $t = -i/2$, called the *Radon-Nikodym point*:

$$\Delta_{\text{RN}}^{1/2} = [D\omega : D\psi]_{-i/2} = \Delta_{\omega|\psi}^{-1/2} \Delta_{\psi}^{1/2}, \quad (42)$$

This corresponds to an “imaginary” Loschmidt echo,¹⁹ probing the spatial rather than the temporal structure of the cocycle:

$$\mathcal{F}(\omega, \psi) = \langle \psi | \Delta_{\text{RN}}^{1/2} | \psi \rangle = \int e^{-\lambda/2} d\mu(\lambda) + \text{corrections}, \quad (43)$$

where $d\mu(\lambda) = c_\lambda \psi(a_\lambda)$ as above, and corrections arise from commuting the modular Hamiltonians and using Baker-Campbell-Hausdorff. Ignoring these corrections,²⁰ this is just the weighted mean eigenvalue of $\Delta_{\text{RN}}^{1/2}$.

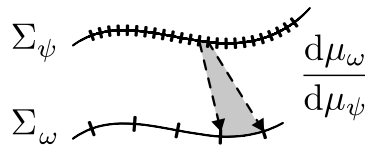
This may seem like an odd place to look, but it allows us to compare *spatial measures* on a spatial slice Σ . More precisely, given a function $f \in C^0(\text{spec}(\Sigma), \mathbb{C}) \cong \Sigma$, the *weight* of the function in a state is simply the evaluation:

$$\psi(f) = \int_{\text{spec}(\Sigma)} f(x) d\mu_\psi(x). \quad (44)$$

As a special case, for the projector Π_R associated to a region $R \subseteq \text{spec}(\Sigma)$, the *volume* of the region is simply

$$\mathcal{V}_\psi(R) = \psi(\Pi_R) = \int_{\text{spec}(\Sigma)} \mathbf{1}_R(x) d\mu_\psi(x) = \int_R d\mu_\psi(x). \quad (45)$$

To convert to the measure in another state ψ , we use the *Radon-Nikodym derivative* $d\mu_\omega/d\mu_\psi$. It is discussed at length in any measure theory text,²¹ but for our



purposes, simply converts between measures:

$$\omega(f) = \int_{\text{spec}(\Sigma)} f(x) d\mu_\omega(x) = \int_{\text{spec}(\Sigma)} f(x) \left. \frac{d\mu_\omega}{d\mu_\psi} \right|_x d\mu_\psi(x). \quad (46)$$

Connes showed²² that, on a commutative subalgebra, the cocycle (42) is precisely the square root of the Radon-Nikodym derivative, and hence the volume in state $|\psi\rangle$ is related to $|\omega\rangle$ via

$$\omega(f) = \langle \psi | \Delta_{\text{RN}}^{1/2} f \Delta_{\text{RN}}^{1/2} | \psi \rangle \implies \mathcal{V}_\omega(R) = \langle \psi | \Delta_{\text{RN}}^{1/2} \Pi_R \Delta_{\text{RN}}^{1/2} | \psi \rangle. \quad (47)$$

This motivates us to study the overlap of this modified Radon-Nikodym state $\Delta_{\text{RN}}^{1/2}|\psi\rangle$ with $|\psi\rangle$ itself, which is precisely the expression (43):

$$\mathcal{F}(\omega, \psi) = \langle \psi | \Delta_{\text{RN}}^{1/2} | \psi \rangle = \langle \omega | \Delta_{\text{RN}}^{-1/2} | \omega \rangle, \quad (48)$$

¹⁹ Since $\mathcal{F}(\omega, \psi) = \mathcal{L}_{K_\omega, K_\psi}^\psi(i/2)$.

²⁰ Which are suppressed when the commutator $[K_\omega, K_\psi]$ is much smaller in operator norm than the cocycle Hamiltonian itself.

²¹ For instance *Real and Complex Analysis* (1990), Walter Rudin.

Figure 6: The Radon-Nikodym derivative converts between measures.

²² *Noncommutative Geometry* (1990), Alain Connes. Hence his self-deprecating terminology of *Radon-Nikodym cocycle*.

noting a pleasingly symmetric definition in terms of $|\omega\rangle$ which follows immediately from (47), and is consistent with $\Delta_{\text{RN}}^{-1} = d\mu_\psi/d\mu_\omega$. When the states are equal, $d\mu_\omega/d\mu_\psi = 1$ is identically unity, and hence

$$\mathcal{F}(\omega, \psi) = \langle \psi | \Delta_{\text{RN}}^{1/2} | \psi \rangle = 1. \quad (49)$$

Moreover, since $\langle \psi | \Delta_{\text{RN}} | \psi \rangle = \langle \omega | \omega \rangle = 1$, Cauchy-Schwarz shows that

$$|\mathcal{F}(\omega, \psi)|^2 = |\langle \psi | \Delta_{\text{RN}}^{1/2} | \psi \rangle|^2 \leq 1. \quad (50)$$

Less trivial is the fact that the inequality is *strict* when $|\omega\rangle \neq |\psi\rangle$, just like relative entropy. The imaginary Loschmidt echo $\mathcal{F}(\omega, \psi)$ is more commonly known as the *fidelity* in the quantum information literature,²³ and is a robust measure of the difference between quantum states.

²³ See for instance *Quantum Computation and Quantum Information* (2000), Michael Nielsen and Isaac Chuang.

We find that, in the same way that relative entropy grounds modular time dilation, fidelity is the invariant which grounds modular length contraction; it is invariant under automorphisms that preserve $|\psi\rangle$ for similar reasons. In terms of (43), and focusing on the leading terms in the λ integral,

$$\int e^{-\lambda/2} d\mu(\lambda) \lesssim 1. \quad (51)$$

Interpreting $e^{-\lambda/2}$ as a length contraction factor, it seems that moving rulers shrink on average in the same way that moving clocks slow on average, albeit with slightly different averages. But they are directly connected in the case of a single, perfectly localized eigenvalue $d\mu(\lambda) = \delta(\lambda - \lambda_0) d\lambda$, where we have the striking relation

$$\mathcal{F}(\omega, \psi)^2 = e^{-\lambda_0} = e^{-S(\omega||\psi)}. \quad (52)$$

When there is dispersion, Jensen's inequality is strict and they disagree, with $\mathcal{F} > e^{-S/2}$. Thus, we can think of a well-localized spectrum as a sort of *classical limit* of modular relativity. In contrast, the difference

$$\mathcal{K}(\omega, \psi) = \log[\mathcal{F}(\omega, \psi)^2 e^{S(\omega||\psi)}] \geq 0 \quad (53)$$

represents the sum of higher cumulants.²⁴ We predict generically that $\mathcal{K} > 0$, though we will discuss mechanisms for classical localization of the cocycle spectrum below.

²⁴ $\mathcal{K}$ stands for “kumulant”.

3.3. Phenomenology

Our third question regarded the connection between modular and special relativity. We have been referring loosely to “time dilation”, “length contraction” and “boosts”. To see how this terminology holds up, we can consider the vacuum state on the Rindler wedge. The work of Bisognano and Wichmann²⁵ showed that the modular Hamiltonian can be identified with the boost generator up to a constant, $\Delta_\psi = e^{-2\pi K_{\text{boost}}}$. The *rapidity* η is related to modular time by $\Delta_\psi^{it} = e^{-\eta K_{\text{boost}}}$, and diagonalizes in *null coordinates* $x^\pm = t \mp x$ with

²⁵ “On the duality condition for a Hermitian scalar field” (1975), Bisognano and Wichmann.

$$x^\pm \mapsto e^{\pm\eta} x^\pm. \quad (54)$$

We depict this in Fig. 8. This suggests we identify relative entropy with the average eigenvalue of the “boost” operator connecting states. Since that value scales with rapidity η , we might expect average rapidity in state $|\psi\rangle$ to be proportional to relative entropy:

$$\langle\eta\rangle_\psi \propto S(\omega\|\psi). \quad (55)$$

Since they are tied to scaling in null directions, modular “time dilation” and “length contraction” might better be termed *modular null contraction* and *null dilation*. This identifies modular time as stretched along one null direction (positive rapidity²⁶) and space as squeezed along the other (negative rapidity).

²⁶ Due to positivity of relative entropy.

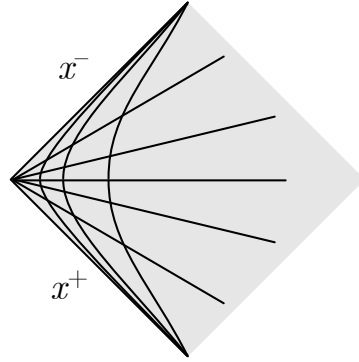


Figure 7: Rindler space, with modular flow given by lines of constant acceleration, and spatial slices fanning out from the origin. The null coordinates $x^\pm$ are indicated.

Why don’t our earlier arguments reproduce the standard time dilation and length contraction factors? The short answer is that the frequencies associated to modular evolution are (in geometric settings) related to boosts, and these act diagonally in the null directions. Similarly, the Radon-Nikodym derivative is the Jacobian of a change of *spacetime* measure, restricted to a spacelike surface. The e^η scaling in modular time induces a $e^{-\eta}$ scaling in modular space, and this shows up at the RN point.²⁷

Although the modular null scalings are not identical to the standard phenomenology, we can still recover the Lorentz factor from kinematics. Recall that γ is related to rapidity by

$$\gamma = \cosh(\eta) = \frac{1}{2}(e^\eta + e^{-\eta}). \quad (56)$$

A moving clock slows by γ , since (setting $x = 0$, i.e. stationary in original frame)

$$t = \frac{1}{2}(x^+ + x^-) \mapsto t' = \frac{1}{2}(e^\eta + e^{-\eta})t = \gamma t. \quad (57)$$

On the other hand, a moving ruler is measured to have simultaneous endpoints:

$$\Delta t \mapsto \Delta t' = \frac{1}{2}(e^\eta + e^{-\eta})t + \frac{1}{2}(e^\eta - e^{-\eta})x = 0. \quad (58)$$

²⁷ To be clear, spacelike surfaces do not contain a null direction; the scaling is cleanest in null coordinates.

We can substitute this into the expression for $\Delta x'$:

$$\begin{aligned}
 \Delta x &= \frac{1}{2}(\Delta x^+ - \Delta x^-) \mapsto \Delta x' = \frac{1}{2}[e^\eta(\Delta t + \Delta x) - e^{-\eta}(\Delta t - \Delta x)] \\
 &= \frac{1}{2}(e^\eta - e^{-\eta})\Delta t + \frac{1}{2}(e^\eta + e^{-\eta})\Delta x \\
 &= \frac{1}{2} \left[\frac{-(e^\eta - e^{-\eta})^2}{(e^\eta + e^{-\eta})} + (e^\eta + e^{-\eta}) \right] \Delta x \\
 &= \frac{1}{2} \left[\frac{4}{(e^\eta + e^{-\eta})} \right] \Delta x = \frac{\Delta x}{\gamma},
 \end{aligned}$$

using the simultaneity condition on the third line. This gives the familiar forms of time dilation and length contraction.

4. Examples

In this section, we explore some simple, tractable examples which ground the identification of relative entropy with (average) rapidity, and shed light on the relative modular spectrum. We first consider a finite-dimensional toy model, then a simple field theory example where we compare the vacuum state to a coherent excited state.

4.1. A toy model

Consider a commutative algebra $\mathfrak{M}$ over a finite-dimensional Hilbert space $\mathcal{H}$, consisting of mixed states diagonal in some common basis:

$$\rho_i = \sum_E p_i(E) |E\rangle\langle E|. \quad (59)$$

Suppose the algebra acts on a Hilbert space $\mathcal{H}$, with commutant²⁸ $\mathfrak{M}'$. The theory of finite-dimensional modular flow tells us that

²⁸ This is the set of operators in $\mathfrak{B}(\mathcal{H})$ which commute with each element of $\mathfrak{M}$, schematically $[\mathfrak{M}, \mathfrak{M}'] = 0$.

$$\Delta_i = \rho_i \otimes \rho_i^{-1}, \quad \Delta_{i|j} = \rho_i \otimes \rho_j^{-1}, \quad [D\rho_i : D\rho_j]_{-i} = \rho_i \rho_j^{-1} \otimes e. \quad (60)$$

Thus, evaluated at a fixed energy E , the relative modular operator is just the likelihood ratio, and the cocycle spectral distribution is the log likelihood:

$$[D\rho_i : D\rho_j]_{-i} = \frac{p_i(E)}{p_j(E)}, \quad \lambda(E) = -\log \Delta_{i|j}(E) = \log \left[\frac{p_j(E)}{p_i(E)} \right]. \quad (61)$$

The relative entropy is the expected log likelihood or KL divergence:

$$S(\rho_i \| \rho_j) = \mathbb{E}_j \left[\log \frac{p_j}{p_i} \right] = \mathbb{E}_j [\lambda(E)]. \quad (62)$$

Thus, $\lambda(E)$ is the spectral density of the cocycle Hamiltonian and $S(\rho_i \| \rho_j)$ is the averaged spectral density.

On the other hand, at the Radon-Nikodym point, $[D\rho_i : D\rho_j]_{-i/2} = \rho_j^{1/2} \rho_i^{-1/2}$, and hence the fidelity is

$$\mathcal{F}(\rho_i, \rho_j) = \text{Tr} [\rho_j [D\rho_i : D\rho_j]_{-i/2}] = \text{Tr} [\sqrt{\rho_i} \sqrt{\rho_j}] = \sum_E \sqrt{p_i(E) p_j(E)}, \quad (63)$$

the classical Bhattacharyya distance. This can also be written

$$\mathcal{F}(\rho_i, \rho_j) = \mathbb{E}_j \left[e^{-\lambda/2} \right] \quad (64)$$

in accord with the integral (43). To explore localization of the spectrum, note that, by Taylor expanding the exponential and the logarithm, we have

$$\log \mathcal{F} = \log \mathbb{E}_j \left[e^{-\lambda/2} \right] = -\frac{1}{2} \mathbb{E}_j[\lambda] + \frac{1}{8} \sigma_j^2[\lambda] - \frac{1}{48} \kappa_3 + \cdots, \quad (65)$$

or equivalently

$$\mathcal{K} = \log(\mathcal{F}^2 e^S) = \frac{1}{4} \sigma^2 + \mathcal{O}(\kappa_3). \quad (66)$$

This shows explicitly how, at least in finite dimensions, $\mathcal{K}$ is indeed a sum of higher cumulants as claimed earlier.

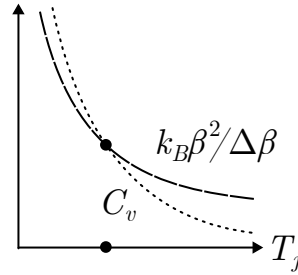


Figure 8: Heat capacity versus inverse temperature. There is an $\mathcal{O}(1)$ crossover point provided $T_i \sim T_j$.

As a special case which illustrates localization neatly, consider thermal density matrices with respect to the same Hamiltonian,

$$\rho_i = \frac{1}{Z(\beta_i)} e^{-\beta_i H}, \quad Z(\beta_i) = \text{Tr}[e^{-\beta_i H}]. \quad (67)$$

A short calculation shows that

$$\sigma_j^2[\lambda] = (\beta_j - \beta_i) \sigma_j^2[E] = (\beta_j - \beta_i) \partial_\beta^2 \log Z(\beta) \Big|_{\beta=\beta_j} = \frac{\Delta \beta \cdot C_v(T_j)}{k_B \beta_j^2}, \quad (68)$$

where C_v is the thermal heat capacity. Thus, the cocycle spectrum is effectively localized (small variance) provided the inverse temperature is much larger than the heat capacity:

$$C_v(T_j) \ll \frac{k_B \beta^2}{\Delta \beta} = \frac{1}{T_j(1 + T_j/T_i)}. \quad (69)$$

For a finite-level system, we typically have²⁹ $C_v(T) \propto T^{-2}$, so localization occurs in the high-temperature limit provide $T_i \sim T_j$. See Fig. 8.

²⁹ This is called the *Schottky anomaly*, and occurs due to finite entropy effects.

4.2. Coherent states

While the toy model illustrates the basic structure, there are tractable quantum field theory examples which furnish more geometric intuition for modular relativity. Consider the right Rindler wedge

$$W = \{(t, x_1, \mathbf{x}) \in \mathbb{R}^{0,d} : x_1 > |t|\}, \quad (70)$$

and the associated algebra of local excitations for a Hermitian field

$$\mathfrak{A}(W) = \{\phi(x) : x \in W\}. \quad (71)$$

If $|\psi\rangle$ is the Minkowski vacuum (the unique state invariant under Poincaré transformations) restricted to W , then by the Reeh-Schlieder theorem³⁰ it is cyclic and separating. Bisognano and Wichmann proved that the modular Hamiltonian is precisely the *boost generator* up to a factor of 2π :

$$K_\psi = 2\pi K_{\text{boost}}, \quad (72)$$

with rapidity $\eta = 2\pi t$. A localized coherent excitation is produced by a Weyl unitary $W(f)$ acting on the vacuum, $|\omega_f\rangle = W(f)|\psi\rangle$; this is also cyclic and separating. Assume for simplicity that the smearing function f is localized to the right wedge. Then Casini, Grillo and Pontello proved that³¹

$$S(\omega_f|\psi) = 2\pi \langle f, k_1 f \rangle_{\mathcal{H}_{1P}}, \quad (73)$$

where k_1 is the one-particle boost generator and $\mathcal{H}_{1P}$ is the one-particle Hilbert space.³² If we expand the spectrum of the one-particle boost generator as

$$k_1 = \int \lambda \, d\zeta(\lambda) \quad (75)$$

and substitute into the relative entropy, we obtain

$$S(\omega_f|\psi) = 2\pi \langle f, f \rangle \int \lambda \, d\mu(\lambda), \quad d\mu(\lambda) = \frac{\langle f, d\zeta(\lambda) f \rangle}{\langle f, f \rangle}. \quad (76)$$

Again, we have a direct connection to the spectrum of boosts, up to a prefactor $2\pi \langle f, f \rangle$. If we set $d\mu(\lambda) = \delta(\lambda - \lambda_0) d\lambda$, we find $S(\omega_f|\psi) = 2\pi \langle f, f \rangle \lambda_0$.

Casini, Grillo and Pontello also prove that the Connes cocycle takes the form

$$[D\omega_f : D\psi]_t = W(f) \sigma_\psi^t[W(f)^*] = W(f) W(f_t), \quad (77)$$

where f_t is a boosted smearing function. Taking the vacuum expectation, Cadamuro and Del Vecchio compute the closed form³³

$$\left| \langle \psi | [D\omega_f : D\psi]_{-i/2} | \psi \rangle \right|^2 = \exp \left[-\langle \Delta_\psi^{1/2} f - f, \Delta_\psi^{1/2} f - f \rangle \right]. \quad (78)$$

Specializing to the Rindler wedge, we have $\Delta_\psi^{1/2} = e^{-\pi k_1}$ and for a small, localized eigenpacket ($\Delta_\psi^{1/2} - e \approx -\pi k_1$) we get

$$\left| \langle \psi | [D\omega_f : D\psi]_{-i/2} | \psi \rangle \right|^2 \approx e^{-2\pi \langle f, f \rangle \lambda_0}, \quad (79)$$

which precisely agrees with e^{-S} , as we expect for the well-localized limit. Note that for this to occur, we need localization at small eigenvalues, otherwise corrections (corresponding to higher terms in the λ -integral (43)) spoil the agreement. Thus, we predict nonzero $\mathcal{K}$ at high boost.

³⁰ "Bemerkungen zur Unitäräquivalenz von Lorentzinvarianten Feldern" (1961), Helmut Reeh and Siegfried Schlieder. For a technical definition of the vacuum, see "An algebraic approach to quantum field theory" (1964), Rudolf Haag and Daniel Kastler.

³¹ "Relative entropy for coherent states from Araki formula" (2019), Horacio Casini, Sergio Grillo, and Diego Pontello.

³² We omit the subscript from now on.

³³ "Relative entropy of coherent states on general CCR algebras" (2022), Daniela Cadamuro and Simone Del Vecchio.

5. Open questions

In this section, we explore some loose threads. First, there is the question of localization of the cocycle spectrum. We identified two regimes where localization held: the high-temperature limit of a finite-dimensional system; and a weak, narrowly concentrated boost for coherent excitations in Rindler space. In these cases, the kumulant function $\mathcal{K} \approx 0$ and the fidelity is directly related to the relative entropy. There are two corrections to this picture: higher cumulants in the modular spectrum, and Baker-Campbell-Hausdorff corrections to the Radon-Nikodym density. These can be viewed as stochastic and quantum effects respectively.

If classicality of spacetime (the familiar duality of length contraction and time dilation) is indeed due to localization, this begs the question:

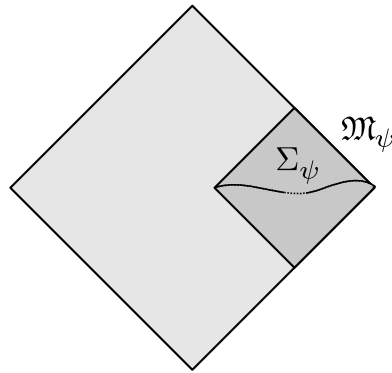
How does classical spacetime suppress corrections to modular localization?

One possibility is that the relative modular Hamiltonian acts as a “pointer” basis for decoherence,³⁴ but this would only suppress quantum corrections, not stochastic, unless each branch of the wavefunction implements a single classical value of λ . This requires a careful census of the degrees of freedom in the model and the roles they play.

Despite its importance, we have remained noncommittal about the nature of the spatial algebra Σ_ψ and the accessible subalgebra $\mathfrak{M}_\psi$. In addition to being a masa of $\mathfrak{M}_\psi$, we could define Σ_ψ generate $\mathfrak{M}_\psi$ in the sense that it is the image of Σ_ψ under modular flow. Mathematically, we can use von Neumann’s bicommutant theorem to pluck out the smallest von Neumann algebra containing this image, and identify with $\mathfrak{M}_\psi$:

$$\mathfrak{M}_\psi = [\sigma_\psi^\mathbb{R}(\Sigma_\psi)]''. \quad (80)$$

The moral is that Σ_ψ shouldn’t have any “holes” that completion into a von



³⁴ “Decoherence, einselection, and the quantum origins of the classical” (2001), Wojciech Zurek.

Figure 9: A situation forbidden by our proposal: Σ_ψ has a (dotted) gap in the von Neumann algebra $\mathfrak{M}_\psi$ it generates. Equivalently, it is not a masa in $\mathfrak{M}_\psi$.

Neumann algebra fills in. This is a start, but can we constrain the global behaviour more strongly? In other words:

What is the global nature of the spatial system Σ_ψ associated to an observer $|\psi\rangle$?

The modular length contraction discussed is *local*, so it is not particularly helpful. One promising direction involves *conditional expectation*,³⁵ channels $\mathbb{E}_\psi$

³⁵ “Conditional expectations in von Neumann algebras” (1972), Masamichi Takesaki.

which collapse operators onto a commutative subalgebra Σ_ψ and which preserve $|\psi\rangle$. They also connect to the rich topic of error correction, but we leave further exploration to future work.

Another possible type of constraint is *thermodynamic* in origin. A large part of Connes and Rovelli’s motivation for using modular time was its thermal nature, since each state obeys the Kubo-Martin-Schwinger³⁶ condition

$$f(t) = \langle a\psi | \Delta_\psi^{it} | b\psi \rangle = \psi(a\sigma_\psi^t(b)), \quad f(t-i) = \psi(\sigma_\psi^t(b)a) \quad (81)$$

under its own modular flow. In words, they are equilibrium states of the flow they generate. This is almost too strong; each observer appears to be in equilibrium by definition, but it is precisely the *non-equilibrium* effects which seem to account for the arrow of time. This suggests a possibility:

Does modular relativity encode non-equilibrium interactions between observers?

Observers may be KMS with respect to their own flow, but they are never KMS under the modular flow of another state. Comparing KMS flows may be the natural way to link thermodynamics to modular time.

Another natural place to seek “interactions” is the interplay between relative entropy and fidelity is. Relative entropy governs boosts, while fidelity controls the Radon-Nikodym Jacobian. Relative entropy also governs the well-known Fisher information metric for quantum and classical states.³⁷ Given that observers *are* states in our formulation, there is an induced metric on the state space of $\mathfrak{M}$, which must be connected to the emergent geometry:

How is the Fisher information metric connected to modular “spacetime”?

Fidelity governs the “unitary” geometry of state space, so the role it plays here (particularly for nonzero kumulant) is crucial to understand better.

Finally, if we are consider the geometry associated with multiple observers, we are in effect discuss *local* Lorentz symmetry, which takes us into the realm of general relativity. Perhaps stochastic corrections in the fidelity encode curvature,³⁸ but this is optimistic speculation for now. The idea of knitting together local modular effects parallels the development of *algebraic quantum field theory*, which generalizes a single von Neumann algebra to a net of local algebras. Hence, we ask the question:

Does modular relativity for a net of local algebras lead to general relativity?

This is closely related to Borchers’ program for recovering Lorentz symmetry from modular flow,³⁹. Given that observers may exhibit nontrivial thermal physics, it’s also natural to wonder how this local story ties into proposals⁴⁰ relating general relativity to thermodynamics. We leave these and other problems to future work.

6. Conclusion

Einstein’s late suspicion was not merely that the continuum is “too big,” but that there is a structural mismatch with what quantum theory demands: finite information about finite systems, encoded algebraically rather than geometrically. In

³⁶ “Statistical-Mechanical Theory of Irreversible Processes I” (1957), Ryogo Kubo, and “Theory of Many-Particle Systems I” (1959), Paul Martin and Julian Schwinger.

³⁷ See e.g. “Statistical distance and the geometry of quantum states” (2001), Samuel Braunstein and Carlton Caves.

³⁸ Or *kurvature*?

³⁹ *On Revolutionizing of Quantum Field Theory with Tomita’s Modular Theory* (1999), H.J. Borchers.

⁴⁰ For instance “Thermodynamics of Spacetime: The Einstein Equation of State” (2001), Ted Jacobson.

the opening vignette, we imagined him drifting—perhaps unwillingly—toward von Neumann’s “rings of operators,” not as a mathematician chasing abstraction, but as a physicist chasing a language in which reality is what can be acted on and measured. This essay took that premise seriously: start with the algebra of operations $\mathfrak{M}$, and ask whether the familiar trappings of spacetime can be recovered as *derived* structures rather than structural assumptions.

What emerges is a remarkably tight triad. Space appears as a (possibly observer-dependent) “spatial” subalgebra whose spectrum behaves like a coordinate space, though one that naturally trades point-locality for projection-locality. Time is not a Newtonian absolute, but the intrinsic one-parameter automorphism group σ_ψ^t associated to a state $|\psi\rangle$ via Tomita–Takesaki theory. And finally, observers are not some external appendage, but the constituents of Hilbert space itself, with relations between observers encoded by the relative modular operator and Connes’ cocycle, the “bridge” that allows to traverse between different modular worlds.

Once one admits observers-as-states, familiar question of relativistic phenomenology can be approached via information theory. The cocycle evaluated near $t = 0$ produces relative entropy, which we interpret as an intrinsic “clock comparison”: a modular analogue of time dilation grounded not in the invariance of c , but in the invariance of *distinguishability* under symmetries of the reference state. Passing from real time to the Radon–Nikodym point $t = -i/2$ produces fidelity, which we read as the spatial analogue: a Jacobian-like factor converting measures on (commutative) spatial slices. In the limit of a well-localized cocycle spectrum, these invariants lock together ($\mathcal{F}^2 \approx e^{-S}$), while dispersion in the relative modular spectrum produces a genuinely new correction term $\mathcal{K}$ —a quantitative reminder that “classical kinematics” is the special case where the cocycle spectrum behaves like a single rapidity.

The Rindler example shows that this is not merely metaphor. In the wedge algebra, Bisognano–Wichmann identifies the modular Hamiltonian of the vacuum with the Lorentz boost generator (up to 2π), so modular time becomes rapidity, and the “dilation/contraction” language becomes literally geometric, i.e. diagonal in null coordinates. The coherent-state comparison then supplies a second anchor: relative entropy and the cocycle admit closed expressions in terms of the one-particle boost generator and boosted smearings, making the information-theoretic invariants computable and the notion of localization concrete. In this case, special relativity emerges as the geometric avatar of the underlying modular structure.

So, what becomes of Einstein’s pessimism? If spacetime is indeed the wrong primitive, then the “purely algebraic theory” he regarded as unknowable may not look like a discrete lattice, or a combinatorial replacement for $\mathbb{R}^{1,3}$, but rather like a theory in which spacetime is the macroscopic organization of information. This flavour of idea is by no means new.⁴¹ But our approach has the unusual distinction of throwing away spacetime-as-primitive altogether: no boundary theory, AQFT, or discretization of spacetime, just a crystalline structure of things one can do or see, and states to do and see them. In that light, the sea-image from the introduction is apt: one does not break the continuum; one

⁴¹ See for instance “Building up spacetime with quantum entanglement” (2010), Mark Van Raamsdonk; and “Cool horizons for entangled black holes” (2013), Juan Maldacena, Leonard Susskind.

quietly surrounds it, until geometry is no longer foundational but emergent.

The open questions in §5 are therefore not epilogues but the real beginning: what selects the “right” Σ_ψ for an observer; how classical localization suppresses cumulants and Baker-Campbell-Hausdorff corrections; how the Fisher metric on state space becomes the seed of spacetime geometry; and whether a net of local algebras forces a general-relativistic analogue of local Lorentz symmetry. If modular relativity is more than a parable, its next tests will come precisely where Einstein’s instincts were sharpest: in the tension between locality, covariance, and thermodynamics, where “time” is not a background parameter but something an observer extracts from the algebra of the world.

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